

## Class 28: Eigenvectors and Eigenvalues

**Definition:** If  $A$  is a square matrix, we define an eigenvector for  $A$  to be \_\_\_\_\_

Example in  $\mathbf{R}^3$ . 
$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 3 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} =$$

Show that  $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$  is also an e-vector for  $A$ . What is the corresponding e-value?

Example: Given  $\lambda = 4$  is an eigenvalue for  $A = \begin{bmatrix} 1 & 1 \\ 6 & 2 \end{bmatrix}$  find an eigenvector.

**Theorem:** Let  $A$  be any  $n \times n$  matrix. The set of all vectors satisfying  $A\mathbf{x} = \lambda\mathbf{x}$  is a sub space of  $\mathbf{R}^n$  called the eigenspace of  $A$  corresponding to  $\lambda$ .

Proof:

**Practice:** Find all eigenvalues for a diagonal matrix.

**Question:** How do you find all eigenvalues for a square matrix  $A$ ?

\* $\lambda$  is an eigenvalue for  $A$  if and only if  $\det(A - \lambda I) = 0$ .

**Practice:** Find all the eigenvalues for the matrix  $A = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix}$

E-values for  $A$  are the solutions to the characteristic equation for  $A$ .

Find a basis for the eigenspace of  $A$  corresponding to eigenvalues 4 and 1.

Remember this is just finding a basis for the null space of the matrix  $A - 4I$  and matrix  $A - I$ .